# Supplementary material for: The Boussinesq–compressible gap in steady two-dimensional buoyancy-driven enclosure convection


## Abstract

This supplement collects the tables that support the main text but are not needed in the flow of the argument: the asymptotic predictions and their numerical confirmation, the two-parameter fits, the exact single-$\varepsilon$ transfer matrix, the learning curve, the public-data splits, the prefactor–velocity law, and the mapping from scripts to reported results. Table and section numbers in cross-references refer to the main text.


The runs behind every number in the main text are: 600 main solutions (`run_batch.sh`), 100 paired (`run_paired.sh`), 27 at $\varepsilon \to 0$ (`run_eps0.sh`), 104 at $Pr = 4.38$ (`run_water.sh`); `run_cpu_queue.sh` adds the 2:1 geometry and the tight-tolerance reruns.

**Table S1:** The five testable consequences of the expansion and their measured values.

| prediction | measurement |
|---|---|
| $R \to 0$ as $\varepsilon \to 0$ | intercept $+0.0003 \pm 0.0047$ ($u_x$); $0.06\sigma$ from zero |
| $R \propto \varepsilon$ | $m = 1.023 \pm 0.018$ / $0.995 \pm 0.019$ / $1.009 \pm 0.015$ ($u_x/u_y/p$), window-averaged over the two decades; the residual is the $\varepsilon^2$ term (Sec. 5.1) |
| the shape of $D$ does not rotate with $\varepsilon$ | $\cos(D(\varepsilon), D(1.2)) = 0.95$–$1.00$, four channels $\times$ five values of $\varepsilon$ |
| $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$ | $\mathrm{div}(Y)$ vs $\varepsilon \nabla \cdot (\theta \boldsymbol{u}_X)$: cosine $+0.994$ for $\varepsilon < 0.6$, $+0.965$ overall, amplitudes $2.26 \times 10^{-2}$ vs $2.10 \times 10^{-2}$ |
| the expansion converges fast | per configuration, $\varepsilon^1$ alone carries 86% of the correction energy, $\varepsilon^1 + \varepsilon^2$ carries 99.0%, $\varepsilon^3$ carries 99.9% |

**Table S2:** Two-parameter refinement, $R = A\varepsilon^m Ra^q$. The $Ra$ trend for $u_y$ and $\theta$ reproduces in the 2:1 geometry.

| channel | unit cavity ($n = 600$) $m$ / $q$ / $R^2$ | aspect 2:1 ($n = 104$) $m$ / $q$ / $R^2$ |
|---|---|---|
| $u_x$ | 1.081 / $+0.003 \pm 0.006$ / 0.74 | 1.071 / $-0.143 \pm 0.019$ / 0.71 |
| $u_y$ | 1.016 / $-0.083 \pm 0.005$ / 0.77 | 1.146 / $-0.077 \pm 0.019$ / 0.69 |
| $\theta$ | 1.120 / $+0.514 \pm 0.015$ / 0.72 | 0.989 / $+0.447 \pm 0.045$ / 0.55 |
| $p$ | 1.127 / $-0.029 \pm 0.005$ / 0.83 | 1.057 / $-0.154 \pm 0.017$ / 0.74 |

**Table S3:** Exact transfer error $R(u_x)$ when a field computed at one $\varepsilon$ (rows) is applied at another (columns).

| $R(u_x)$ taken from ↓ / applied at → | 0.4 | 0.8 | 1.2 | 1.6 | 1.9 |
|---|---|---|---|---|---|
| $\varepsilon = 0.4$ | 0 | 0.067 | 0.143 | 0.235 | 0.353 |
| $\varepsilon = 0.8$ | 0.068 | 0 | 0.078 | 0.183 | 0.305 |
| $\varepsilon = 1.2$ | 0.149 | 0.081 | 0 | 0.110 | 0.243 |
| $\varepsilon = 1.6$ | 0.265 | 0.199 | 0.112 | 0 | 0.140 |
| $\varepsilon = 1.9$ | 0.402 | 0.328 | 0.251 | 0.142 | 0 |

**Table S4:** Learning curve on the fixed interpolation test set (200 samples), mean of three seeds. Entries are $u_x/u_y/\theta/p$. P3 keeps $\varepsilon$ outside the network entirely.

| $n$ | B: $\varepsilon$ as input | P3: predict $P(X)$, $\varepsilon$ outside |
|---|---|---|
| 0 (do nothing) | 0.2203/0.2683/0.0381/0.2386 | same |
| 50 | 0.1831/0.1946/0.0269/0.1838 | **0.1308/0.1364/0.0195/0.1501** |
| 100 | 0.1280/0.1422/0.0199/0.1348 | **0.0990/0.1022/0.0136/0.1150** |
| 200 | 0.0974/0.1070/0.0147/0.0998 | **0.0827/0.0866/0.0114**/0.1005 |
| 400 | **0.0674/0.0714**/0.0097/**0.0672** | 0.0735/0.0743/0.0093/0.0896 |

**Table S5:** Median relative correction on all five public splits, at $\varepsilon \equiv 1.2$.

| split | $n$ | $u_x$ | $u_y$ | $\theta$ | $p$ |
|---|---|---|---|---|---|
| train (random walls) | 4000 | 0.1797 | 0.2435 | 0.0228 | 0.1858 |
| test_in_distribution | 200 | 0.1822 | 0.2388 | 0.0220 | 0.1863 |
| test_ood (harder wall dials) | 200 | 0.1567 | 0.2157 | 0.0214 | 0.1671 |
| train_legacy (classical cavity) | 5000 | 0.2988 | 0.3362 | 0.0926 | 0.1989 |
| test_legacy (classical cavity) | 200 | 0.2984 | 0.3390 | 0.0836 | 0.1983 |

**Table S6:** The $\theta$-channel prefactor is the flow strength: $R_\theta/\varepsilon = c\hat{u}^a$ on six data sets from two independent generators.

| data set | $n$ | $R_\theta/\varepsilon = c\hat{u}^a$ | $R^2$ |
|---|---|---|---|
| ours, random walls, $Pr = 0.71$ | 600 | $a = 1.864 \pm 0.026$, $c = 0.544$ | 0.90 |
| public train, random walls | 4000 | $a = 1.902 \pm 0.010$, $c = 0.602$ | 0.91 |
| public test_in_distribution | 200 | $a = 1.977 \pm 0.041$, $c = 0.691$ | 0.92 |
| public test_ood | 200 | $a = 1.931 \pm 0.041$, $c = 0.561$ | 0.92 |
| public train_legacy, classical cavity | 5000 | $a = 2.360 \pm 0.002$, $c = 2.063$ | **0.995** |
| public test_legacy, classical cavity | 200 | $a = 2.368 \pm 0.011$, $c = 2.095$ | **0.996** |

**Table S7:** Scripts and the results they produce.

| script | what it produces |
|---|---|
| `exp_eps_law.py` | the 627-sample law (Sec. 5.1) |
| `exp_grid128.py` | the $64^2$ vs $128^2$ mesh check at $Ra = 10^4$ and $Ra = 10^6$ (Sec. 5.1) |
| `repack_p_fixed.py` | the corrected pressure channel (Sec. 5.2) |
| `exp_paired_final.py` | the $\varepsilon$-blind floor and shape invariance (Sec. 5.3) |
| `exp_modes.py` | the $\varepsilon^k$ mode expansion (Sec. 3) |
| `exp_chi.py` | the $\chi$ explanation for $\theta$ (Sec. 5.1) |
| `harness.py`, `exp_ml_final.py` | the unified surrogate ablation: A/B/P1/P2/P3, three seeds (Secs. 5.4–5.6) |
| `fno_lib.py`, `exp_fno_final.py` | the FNO baseline under the same protocol (Sec. 5.5) |
| `exp_lc_final.py` | learning curves (Sec. 5.6) |
| `exp_pod.py` | the POD-ROM / linear-surrogate baselines (Sec. 5.4) |
| `exp_direct_linear.py`, `exp_direct_batch.py` | the direct solution of the linearised $O(\varepsilon)$ system on all 20 configurations (Sec. 5.4) |
| `exp_direct_eps0.py` | the same one-Newton-step solve on the 27 low-contrast samples (Sec. 5.4) |
| `exp_heldout_loweps.py`, `run_loweps.sh` | the 38 held-out solutions at $\varepsilon = 0.1, 0.2$ and the extrapolated-$P$ check (Sec. 5.4) |
| `exp_direct_low.py`, `exp_fixed_flow_eps.py` | one-Newton-step solves at $\varepsilon = 0.1, 0.2$ on the paired flows and the fixed-flow $\varepsilon$-drift (Sec. 5.4) |
| `exp_cluster_boot.py` | family-cluster bootstrap of the exponent (Sec. 5.1) |
| `exp_p2_fixed.py`, `exp_prior_ablation.py`, `exp_fno.py`, `exp_learning_curve.py` | first-generation runs of the same experiments (superseded by the three scripts above; their JSON outputs were deleted as stale) |
| `exp_public_transfer.py` | external validation and transfer (Secs. 5.7–5.9) |
| `exp_aspect2.py` | the 2:1 geometry (Sec. 5.10) |
| `exp_tight.py`, `verify_noise.py`, `exp_final_numbers.py` | noise control and timing (Secs. 5.11–5.12) |
| `exp_water.py` | the strictly paired $Pr = 4.38$ fluid study (Sec. 5.13) |
| `exp_perturbation.py` | the first-order perturbation baseline on the $\varepsilon$ scan (Sec. 5.4) |
| `exp_linear_response.py` | the linear-response and secant baselines from the paired scan (Sec. 5.4) |
| `exp_second_order.py` | the two-term $P + \varepsilon Q$ fit from the paired scan (Sec. 5.4) |
| `exp_dnu_eps.py` | the integral $Nu$ gap versus $\varepsilon$ (Sec. 5.8) |
| `exp_film.py` | the FiLM-conditioned baseline (Sec. 5.5) |
| `exp_3d_law.py`, `exp_3d_summary.py`, `make_fig9.py`, `exp_3d_rom_eps.py` | the fifteen-sample, five-contrast three-dimensional check and the POD-ROM baseline (Sec. 5.14) |
| `verify_conventions.py` | the 38 400-value convention audit against the published `relL2` columns (Sec. 5.7) |
| `exp_public_splits.py`, `exp_legacy5000.py`, `exp_uhat_law.py` | the five public splits and the $\hat{u}$ law (Sec. 5.7) |
| `exp_transfer_uhat.py` | the $\hat{u}$ transfer criterion on the public splits (Sec. 5.9) |
| `audit_v2.py` | robustness checks: $m$ vs fitting window, $\kappa$, pressure-pair conventions, $\hat{u}$ medians, the 2:1 $\sigma$ distances |
| `make_figs.py`, `make_figs2.py` | every figure, from the JSON results |
| `make_tables.py` | regenerates every table from the JSON results |